

Physics

PART – I

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

- A telescope has an objective of focal length 100 cm and an eyepiece of focal length 2.5 cm. If you look into the objective (that is, into the wrong end) of this telescope, you will see distant objects reduced in size. By what factor will the angular size of objects be reduced?

(A) 40
(B) 10
(C) 20
(D) 4
- Consider a body moving along a circular path of radius R. Given some statements:-

(a) The acceleration of the body is directed towards the centre
(b) The acceleration of the body is directed towards the tangent.
(c) One of the forces acting on the body must be variable in nature.
(d) The radial acceleration of the body is non-zero.

Which of the statements are always true?

(A) a,b,d
(B) a,b,c,d
(C) b,c,d
(D) c,d
- Assume that you are facing south direction at a time when the sun is just setting. The scattered light that comes to you is polarized in a plane which is

(A) Horizontal with electric field along vertical direction.
(B) Vertical with electric field along east west direction.
(C) Vertical with electric field along vertical direction.
(D) Horizontal with electric field along east west direction.
- Light with wavelength 500 nm falls on a slit of width $10\text{ }\mu\text{m}$ at an angle 30° to its normal. Find the angular position of the first minima located on both sides of the central Fraunhofer maximum:-

(A) 33° and 27°
(B) 25° and 35°
(C) 36° and 24°
(D) 28° and 32°
- Let A_n be the area enclosed by the n^{th} orbit in a hydrogen atom. The graph of $\ln(A_n / A_1)$ against $\ln(n)$

(A) Will not pass through the origin
(B) Will have certain points lying on a straight line with slope 4
(C) Will be a monotonically increasing non linear curve
(D) Will be a circle

6. Diatomic gas is used in Carnot heat engine. If efficiency of given Carnot heat engine is 80%, the find the ratio of initial volume to final volume of gas during adiabatic expansion:

- (A) $\left(\frac{1}{5}\right)^{3/2}$
 (B) $\left(\frac{1}{3}\right)^{5/2}$
 (C) $\left(\frac{1}{5}\right)^{5/2}$
 (D) $\left(\frac{1}{5}\right)^{2/5}$

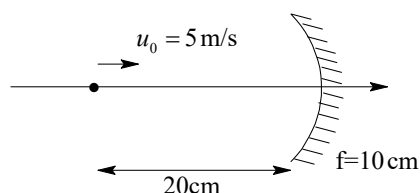
7. A spring mass system executes damped harmonic oscillations given by the equation

$$y = Ae^{-\frac{bt}{2m}} \sin(\omega't + \phi)$$

Where the symbols have their usual meanings. If a 2kg mass (m) is attached to a spring of force constant (K) 1250 N/m, the period of the oscillation is $(\pi/12)s$. The damping constant 'b' has the value.

- (A) 9.8 kg/s
 (B) 2.8 kg/s
 (C) 98 kg/s
 (D) 28 kg/s
8. When a screw gauge is completely closed, zero of circular scale is 7 divisions above the reference line of graduation when the zero of main scale is not seen. If LC of screw gauge is 10^{-3} cm, the zero error is
 (A) -7×10^{-3} cm
 (B) $+7 \times 10^{-3}$ cm
 (C) -0.007 mm
 (D) $+0.007$ mm

9. An object moves with a uniform velocity u_0 along the axis of a concave spherical mirror. Consider the instant shown in diagram, object is moving with $u_0 = 5$ m/s and $f = -10$ cm. If object is at the centre of curvature at this instant then the magnitude of acceleration of image at this instant is a m/s^2 .



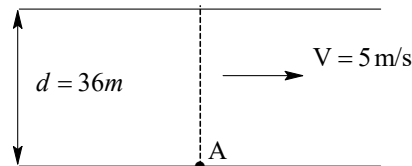
Find a.

- (A) 10
 (B) 5
 (C) 25
 (D) None of these

10. The coil in a moving coil Galvanometer experiences torque proportional to current passed through it. If a steady current i is passed through it the deflection of the coil is found to be 90° . Now the steady current is switched off and a charge Q is suddenly passed through the coil. If the coil has N turns of area A each and its moment of inertia is I about the axis it is going to rotate then the maximum angle through which it deflects upon passing Q is:

- (A) $Q\sqrt{\frac{\pi NAB}{2iI}}$
 (B) $\frac{\pi Q}{2}\sqrt{\frac{NAB}{iI}}$
 (C) $\pi Q\sqrt{\frac{NAB}{2iI}}$
 (D) $Q\sqrt{\frac{\pi NAB}{iI}}$

11. A boat starts from one bank of the river from point A to reach other by crossing width d of the river with minimum drift. Speed of a boat with respect to river is 4 m/s and speed of river with respect to ground is 5 m/s. Value of $d = 36$ m, then



- (A) Minimum drift will be zero.
 (B) Minimum drift will be 27m.
 (C) Time taken for minimum drift will be 20 sec.
 (D) Angle at which boat will be steered from river flow direction for minimum drift is 127° .
12. A pipe of length 1.5 m closed at one end is filled with gas and resonates in its fundamental mode with a tuning fork. Another open organ pipe of same dimensions filled with air resonates in its fundamental mode with same tuning fork. If experiment is performed at 30°C (speed of sound in air is 360 m/sec at 30°C), the speed of sound at 0°C in gas is (Neglect end correction):
- (A) 580 m/sec.
 (B) 683 m/sec.
 (C) 880 m/sec.
 (D) 743 m/sec.
13. A uniform string suspended vertically. A transverse pulse is created at the top most of the string. Then
- (A) Speed of pulse remains constant.
 (B) The speed of the pulse decreases with constant rate as pulse moves downward.
 (C) The speed of the pulse decreases with increasing rate as pulse moves downward.
 (D) The speed of the pulse increases with constant rate as pulse moves downward
14. An N-P-N transistor in a common emitter mode is used as a simple voltage amplifier with a collector current of 4mA. The terminal of a 8V battery is connected to the collector through a load resistance R_L and to the base through a resistance R_B . The collector emitter voltage $V_{CE} = 4\text{V}$, base-emitter voltage $V_{BE} = 0.6\text{V}$ and base current amplification factor $\beta_{DC} = 100$. Calculate the value of R_L and R_B .
- (A) $R_L = 1\text{k}\Omega$, $R_B = 185\text{k}\Omega$
 (B) $R_L = 2\text{k}\Omega$, $R_B = 150\text{k}\Omega$
 (C) $R_L = 1\text{k}\Omega$, $R_B = 240\text{k}\Omega$
 (D) $R_L = 3\text{k}\Omega$, $R_B = 185\text{k}\Omega$

15. In a single slit diffraction pattern, if first maxima of light of wavelength λ coincides with first minima of yellow light (540 nm), then λ is:
(A) 270 nm
(B) 360 nm
(C) 540 nm
(D) 810 nm
16. If there is an error of 1% in calculation of mass of disc & 1.5 % error in radius, then % error in moment of inertia about an axis tangent to disc is:
(A) 2.5%
(B) 4%
(C) 3.5%
(D) 5%
17. To convert galvanometer into ammeter, shunt of 0.01Ω is used. Resistance of galvanometer coil is 50Ω and its maximum deflection current is 20 mA. Range of ammeter is:
(A) 0-1 A
(B) 0-10 A
(C) 0-100 A
(D) 0-1000 A
18. Unpolarized light falls on two polarizing sheets placed one after another. What should be the angle between pass axis of the sheets so that final transmitted light have one fourth intensity of incident light?
(A) 30°
(B) 0°
(C) 60°
(D) None of these
19. A hydrogen atom at rest emits a photon during its transition from $n = 2$ to $n = 1$. Choose the **INCORRECT** statement.
(A) De-Broglie wavelength of hydrogen atom during recoil will be same as wavelength of emitted photon.
(B) Energy of emitted photon will be less than 10.2eV
(C) Kinetic energy of recoil H-atom will be less than 10.2eV
(D) Kinetic energy of recoil H-atom will be equal to energy of emitted photon
20. A wheel of radius r starts rolling with an angular velocity ω and initial linear velocity V_0 up and inclined smooth plane. The wheel will stop going up in time:
(A) $\frac{V_0}{g \sin \theta}$
(B) $\frac{r\omega}{g \sin \theta}$
(C) $\frac{V_0 - r\omega}{g \sin \theta}$
(D) $\frac{r\omega - V_0}{g \sin \theta}$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A particle executes SHM along a straight line with mean position at $x = 0$, time period 12 second and amplitude 5cm. Find the minimum time (in second) in which particle will move from $x = -3\text{cm}$ to $x = 4\text{cm}$
22. Concrete sphere of radius R has a cavity of radius r which is packed with sawdust. The specific gravities of concrete and sawdust are respectively 2.4 and 0.3. For this sphere to float with its entire volume submerged under water, ratio of mass of concrete to mass of sawdust will be:-

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A paramagnetic sample shows a net magnetization of 8 Am^{-1} when placed in an external magnetic field of 0.6 T at a temperature of 4 K . When the same sample is placed in an external magnetic field of 0.2 T at a temperature of 16 K , the magnetization (in Am^{-1}) is:
24. A particle moving with constant speed turns left by an angle of 74° after travelling every 1 m distance. If it returns back to its starting point after 18 sec, then the magnitude of average acceleration of the particle from its initial position to just before it takes its second turn is 50 k m/s^2 . Find the value of K .
25. The electrostatic flux through a rectangular area of $\frac{7}{188}\text{ m}^2$ lying in a plane $2x + 3y + 6z = 10$ due to an electric field $(8\hat{i} + 6\hat{j} + 10\hat{k})\text{ V/m}$ will be in (V-m)